Mini-Project: Robust Asymptotic Growth under Model Uncertainty

ETH zürich

Mathematics for New Technologies in Finance
Prof. Dr. Josef Teichmann
Student: Lorenzo Barbero
GitHub: github.com/Lorbar8/Math-for-New-Tech-Mini-Project

Part I

Exercise 1. The subset $\Theta_0 \subset \Theta$ is a subclass of admissible trading strategies such that the trading strategy θ_t is obtained as the gradient of a function $\varphi \in \mathcal{D}$. The class $\mathcal{D}$ is a class of smooth functions with sufficient "integrability conditions".

Exercise 2. We consider the log-wealth $\log V_T^{\theta_0} = \log \mathcal{E}(\int_0^T \theta_t^\top dX_t)$. Since X_t is assumed to be an O-U process, it has continuos trajectories. For this scenario, the formula for the stochastic exponential is given by:

$$\mathcal{E}(X) = e^{X_t - \frac{1}{2}[X]_t} \Rightarrow \log V_T^{\theta_0} = \int_0^T \theta_{0,t}^\top dX_t - \frac{1}{2}\left[\int_0^\cdot \theta_{0,t}^\top dX_t\right]_T = \int_0^T \theta_{0,t}^\top dX_t - \frac{1}{2}\int_0^T \theta_{0,t}^\top c\theta_{0,t} dt \tag{1}$$

Since $\theta_0 \in \Theta_0$, we can write it as the gradient of φ:

$$\int_0^T \nabla\varphi^\top(X_t)dX_t - \frac{1}{2}\int_0^T \nabla\varphi^\top(X_t)c\nabla\varphi(X_t)dt \tag{2}$$

We can eliminate the stochastic integral by applying Ito's Lemma to the function $\varphi(X_t)$.Therefore, plugging the result and dividing both sides by T, we get:

$$\frac{1}{T}\log V_T^{\theta_0} = \frac{\varphi(X_T) - \varphi(X_0)}{T} - \frac{1}{2T}\int_0^T [\mathrm{Tr}(c\nabla^2\varphi(X_t) + \nabla\varphi(X_t)^\top c\nabla\varphi(X_t)]dt \tag{3}$$

We can then take the limit and use the ergodic property:

$$g(\theta_0, \mathbb{P}) = \lim_{T\to\infty}\frac{1}{T}\log V_T^{\theta_0} = -\frac{1}{2}\int_E \left[\mathrm{Tr}(c\nabla^2\varphi(x)) + \nabla\varphi(x)^\top c\nabla\varphi(x)\right] p(x)\,dx. \tag{4}$$

And we can notice that the problem, because of the ergodic property, does not depend any longer on the probability law $\mathbb{P}$, and our optimization problem turns into: $\sup_{\theta_0\in\Theta_0} g(\theta_0)$.

Exercise 3. Using the definition of the trace, we get:

$$I = \sum_{i,j}\int_E A_{ij}(x)\frac{\partial_{ij}e^{\varphi(x)}}{e^{\varphi(x)}}dx. \tag{5}$$

Since φ is compactly supported, the boundary terms vanish, and integration by parts yields:

$$\int_E A_{ij}(x)\partial_{ij}e^{\varphi(x)}dx = -\int_E \partial_j A_{ij}(x)\,\partial_i e^{\varphi(x)}dx. \tag{6}$$

Then, we obtain

$$I = -\sum_{i,j}\int_E \partial_j A_{ij}(x)\,\partial_i\varphi(x)\,dx = -\int_E (\mathrm{div}A(x))^\top\nabla\varphi(x)\,dx, \qquad (\mathrm{div}A)_i = \sum_j \partial_j A_{ij}. \tag{7}$$

Exercise 4. Recall the definition of the divergence of a matrix field M:

$$(\mathrm{div}M(x))_i = \sum_{j=1}^d \partial_j M_{ij}(x), \quad \Rightarrow \quad (\mathrm{div}A(x))_i = \sum_{j=1}^d \partial_j\left(p(x)c_{ij}\right) = \sum_{j=1}^d c_{ij}\partial_j p(x). \tag{8}$$

Therefore,

$$A(x)^{-1}\mathrm{div}A(x) = (p(x)c)^{-1}c\nabla p(x) = \frac{c^{-1}c\nabla p(x)}{p(x)} = \frac{\nabla p(x)}{p(x)} = \nabla\log p(x) = -\Sigma^{-1}x. \tag{9}$$

And the last equality is obtained since $p(x)$ is a gaussian density.

Exercise 5. In the previous exercise, we computed the mathematical steps required to transform the problem into an LQ problem. Notice that premultiplying the result in exercise 4 by the matrix A, we get $\mathrm{div}A(x) = -c\Sigma^{-1}xp(x)$. Therefore, plugging this result into the integral from exercise 2, we get:

$$-\int_E (\mathrm{div}A(x))^\top \nabla\varphi(x)\,dx = \int_E x^\top \Sigma^{-1} c\nabla\varphi(x)p(x)\,dx \tag{10}$$

And the growth rate becomes:

$$g(\theta_0, P) = -\frac{1}{2}\int_E \left[x^\top \Sigma^{-1} c\nabla\varphi(x) + \nabla\varphi(x)^\top c\nabla\varphi(x)\right] p(x)\,dx. \tag{11}$$

This is a Linear-Quadratic (LQ) optimization problem since the objective contains a term linear in $y := \nabla\varphi$ and a quadratic term in $\nabla\varphi$: $-\frac{1}{2}\left(x^\top\Sigma^{-1}cy + y^\top cy\right)$. Then, we obtain the FOC and the solution to the problem (after integrating):

$$x\Sigma^{-1}c + 2cy = 0 \;\Rightarrow \nabla\varphi^*(x) = -\frac{1}{2}\Sigma^{-1}x \;\Rightarrow \varphi^*(x) = -\frac{1}{4}x^\top\Sigma^{-1}x. \tag{12}$$

Exercise 6. Therefore, since we are considering trading strategies in Θ_0, the optimal robust trading strategy is $\theta^*_{0,t} = \nabla\varphi^*(X_t) = -\frac{1}{2}\Sigma^{-1}X_t$.

Exercise 7. We have to plug the robust-optimal strategy into the equation 11:

$$g^* = \frac{1}{8}\int_E x^\top\Sigma^{-1}c\Sigma^{-1}xp(x)dx = \frac{1}{8}\mathbb{E}[X^\top BX] = \frac{1}{8}\mathrm{Tr}(B\Sigma) = \frac{\mathrm{Tr}(\Sigma^{-1}c)}{8}, \qquad B := \Sigma^{-1}c\Sigma^{-1} \tag{13}$$

Part II

Exercise 1. We consider a d-dimensional Ornstein-Uhlenbeck process of the form

$$dX_t = -BX_t\,dt + \sigma dW_t, \tag{14}$$

where $B \in \mathbb{R}^{d\times d}$ is the drift matrix and $\sigma \in \mathbb{R}^{d\times d}$ is the diffusion matrix (we assume $\sigma'\sigma = c$).

In the adversarial formulation, the Agent chooses a trading strategy $\theta(X_t)$, while Nature chooses the drift matrix B. The matrix B must induce the ergodic distribution in the long run, and this is done by optimizing across the B matrices that satisfy the Lyapunov equation (computed in the following exercise), which links the drift matrix of an O-U process to the long run covariance of the ergodic distribution.

Exercise 2. The density of a O-U diffusion process satisfies the Fokker-Planck equation

$$\partial_t p_t(x) = \nabla\cdot(Bx\,p(x)) + \frac{1}{2}\mathrm{Tr}(c\nabla^2 p(x)) = 0.$$

And the zero equality comes from the stationarity property.

For the Gaussian density p, we obtained $\nabla p(x) = -\Sigma^{-1}x\,p(x)$, and $\nabla^2 p(x) = \left(\Sigma^{-1}xx^\top\Sigma^{-1} - \Sigma^{-1}\right)p(x)$. Computing the terms in the Fokker-Planck equation we get:

$$0 = \mathrm{Tr}(B) - \frac{1}{2}\mathrm{Tr}(c\Sigma^{-1}) + x^\top\left(\frac{1}{2}\Sigma^{-1}c\Sigma^{-1} - B^\top\Sigma^{-1}\right)x. \tag{15}$$

For this equality to hold for all $x \in \mathbb{R}^d$, the quadratic form must vanish. Hence,

$$\Sigma^{-1}c\Sigma^{-1} = B^\top\Sigma^{-1} + \Sigma^{-1}B \;\Rightarrow\; B\Sigma + \Sigma B^\top = c. \tag{16}$$

Which is the Lyapunov equation.

Exercise 3. To determine the number of degrees of freedom available to Nature, we define $M := B\Sigma$. Then, the Lyapunov equation becomes $M + M^\top = c$. This relation fixes the symmetric part of M, namely $\frac{1}{2}(M + M^\top) = \frac{1}{2}c$. Therefore, the remaining freedom lies entirely in the antisymmetric part of M. Define $K := \frac{1}{2}(M - M^\top)$, so that $K^\top = -K$. Hence, every admissible matrix M can be written as $M = \frac{1}{2}c + K$,, where K is skew-symmetric. Consequently,

$$B = \left(\frac{1}{2}c + K\right)\Sigma^{-1}. \tag{17}$$

The adversary, therefore, optimizes only over skew-symmetric matrices K. To count the number of degrees of freedom, observe that a skew-symmetric matrix satisfies $K_{ij} = -K_{ji}$, $K_{ii} = 0$. Thus, only the upper

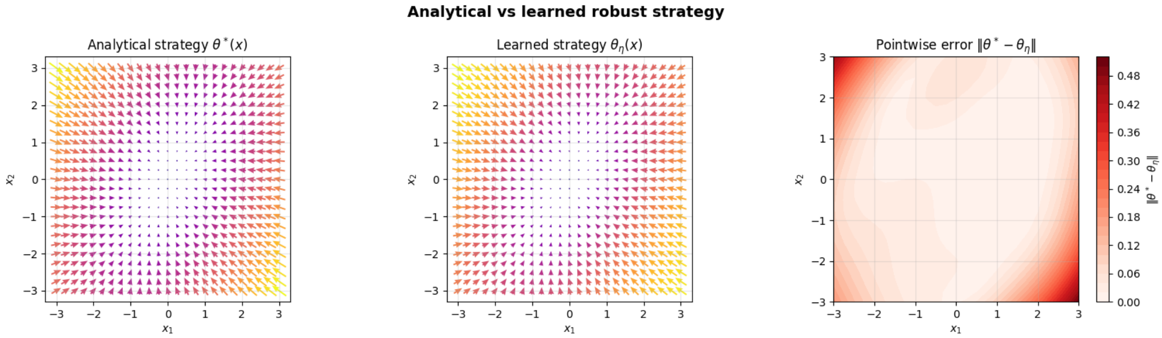

Figure 1: Analyitical and Learned Strategies.

triangular entries matter, and their number is $\frac{d(d-1)}{2}$, which is the number of degrees of freedom of the Fokker-Planck equation.

Exercise 4. We formulate the robust optimization problem as a two-player minimax game. The Agent chooses a trading strategy, while Nature chooses an admissible drift for the Ornstein–Uhlenbeck process. The objective is

$$\max_{\eta} \min_{\xi} \widehat{g}(\eta, \xi),$$

where η denotes the parameters of the Agent and ξ denotes the parameters of Nature.

The Agent is parameterized by a neural network $\theta_\eta : \mathbb{R}^d \to \mathbb{R}^d$. Given the current state X_t, the network outputs the portfolio strategy $\theta_t = \theta_\eta(X_t)$. A natural architecture is a fully connected feedforward neural network with two hidden layers, with tanh activation functions.

Nature chooses the drift of the OU process. Every admissible drift must satisfy the Lyapunov equation. Therefore Nature only needs to parameterize the skew-symmetric matrix K_ξ. We can represent Nature by a small neural network $N_\xi : \mathbb{R}^d \to \mathbb{R}^{d(d-1)/2}$, For fixed Agent and Nature parameters, we simulate paths of the O-U process and we estimate the empirical growth rate:

$$\widehat{g}(\eta, \xi) = \frac{1}{MT} \sum_{m=1}^{M} \sum_{n=0}^{N-1} \left[\theta_\eta(X_{t_n}^{(m)})^\top \Delta X_{t_n}^{(m)} - \frac{1}{2} \theta_\eta(X_{t_n}^{(m)})^\top c \theta_\eta(X_{t_n}^{(m)}) \Delta t \right],$$

where M is the number of simulated paths.

The Agent wants to maximize this quantity, while Nature wants to minimize it. Thus the two losses are

$$L_{\text{Agent}}(\eta, \xi) = -\widehat{g}(\eta, \xi), \qquad L_{\text{Nature}}(\eta, \xi) = \widehat{g}(\eta, \xi).$$

This architecture is appropriate because the OU setting is Markovian, so the trading strategy can be written as a function of the current state X_t.

Exercise 5. At each training iteration, we first construct the matrix B_ξ and simulate paths of the OU process. Using the simulated paths, we estimate the empirical growth rate.

The training is performed by alternating gradient steps. For fixed Agent parameters η, Nature is updated by minimizing the growth rate, while for fixed Nature parameters ξ, the Agent is updated by maximizing the growth rate, equivalently minimizing the negative growth:

$$\xi \leftarrow \xi - \alpha_N \nabla_\xi \widehat{g}(\eta, \xi), \qquad \eta \leftarrow \eta + \alpha_A \nabla_\eta \widehat{g}(\eta, \xi).$$

In practice, this adversarial objective can be unstable because both players change during training. To stabilize the procedure, we use small learning rates, alternate one or a few Nature steps with one Agent step, and regularize Nature's parameters. Gradient clipping can also be used to avoid exploding updates. Training is stopped when the empirical growth rate stabilizes and the learned strategy no longer changes significantly. The resulting Agent network θ_η is then interpreted as the learned robust-optimal trading strategy.

Exercise 6. Figure 1 shows that the learned strategy closely matches the analytically robust-optimal strategy. The empirical growth rate is equal to 0.3516. It quickly stabilizes near the theoretical value $g^* = 0.3395$.

Exercise 7. The numerical results show that the learned strategy closely matches the analytical solution and that the empirical growth rate converges near the theoretically robust-optimal value, despite oscillating across all iterations. This is, of course, suboptimal if one is able to compute the analytical solution, but it shows a valid and flexible approach for more complex settings where explicit solutions are not available.